

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
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Section / Batch:	AJ4ML f2024	Date:	

Instructions to Students

- Answer ALL questions. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Questions	Marks
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis Q1 Q2, Q3	40 60 (2X30)
GRAND TOTAL:		100 Marks

SECTION A – APPLICATION BASED QUESTIONS

Q1.Linear Search – Search for Middle-Late Position Target

Given the unsorted dataset [22, 31, 47, 58, 16, 39, 64], apply Linear Search to find the element 39. Clearly interpret the problem and explain the sequential search process step-by-step. Count the number of comparisons required to locate the target. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when the target appears in the latter half of the dataset.

Q2.Linear Search – Search for Repeated Nonconsecutive Target

Given the unsorted dataset [31, 14, 29, 14, 52, 67, 14], apply Linear Search to find the first occurrence of the element 14. Clearly interpret the problem and explain the sequential search process step-by-step. Count the number of comparisons required before locating the first match. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when repeated target values are scattered across the list.

Code of Conduct

I, P. Maheshwari / 192417294 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (40 Marks)	4	4	4	4	4	40
Q2 (30 Marks)	4	4	4	4	4	30
Q3 (30 Marks)	4	3	3	2	2	22
Overall Total (100)						92

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1.

Linear search - Search for Middle-late position Target!

Problem Interpretation:-

The given unsorted data is:

[22, 31, 47, 58, 16, 39, 64]

The objective is to find the element 39 using linear search. Since the data is unsorted, each element must be checked sequentially from the beginning until the target is found.

Step by-step search Process:-

Comparison	Element checked	Result
1	22	Not found
2	31	Not found
3	47	Not found
4	58	Not found
5	16	Not found
6	39	Found

Number of comparisons

* Total comparisons required = 6.

Time Complexity Analysis:-

- * Best case : $O(1)$ - Target is the first element.
- * Average Case : $O(n)$ - Target is found around the middle of the list.
- * Worst case: $O(n)$ - Target is the last element or not present.

Verification:-

The element 39 is located at index 05.

The search stops immediately after finding the target, confirming the result is correct.

Interpretation:-

Since the target is in the latter half of the dataset, linear search performs ~~many~~ several comparisons before finding it. The execution time increases with the target's position, making linear search less efficient for large unsorted datasets where the target appears near the end.

Linear search - Search for Repeated Non consecutive

Target:-

Problem Interpretation:-

The given unsorted dataset is:

[31, 14, 29, 14, 52, 67, 14]

The objective is to find the first occurrence of the element 14 using linear search. The search stops as soon as the first matching element is found.

Step-by-step Process:-

Comparison	Element checked	Result
1	31	Not found
2	14	Found

Number of comparisons:-

* Total comparisons required = 2

Time complexity Analysis:-

* Best case: $O(1)$ - Target is the first element

* Average case: $O(n)$ - Target is expected around the middle.

* Worst case: $O(n)$ - Target is the last element or absent.

Verification!

The first occurrence of 14 is located at index 1.

Although the value appears again later in the list,

Linear search stops at the first match.

Interpretation!

When repeated target values are scattered across an unsorted list, Linear search efficiently returns the first occurrence without checking the remaining

elements. However, if all occurrences must be found,

the algorithm must continue scanning the entire list, resulting in $O(n)$ time complexity.